MarinOS Foundations Guide

A short guide to the architecture and working methods used by MarinOS.

What is MarinOS?

MarinOS is a shared approach for building small County web applications that are consistent, accessible, secure, and maintainable.

It combines reusable interface components, technical standards, documented practices, and a common delivery model. Each application remains focused on a specific business need.

!**Canonical content flows to reading, quiz, and slide experiences**
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Core principles

MarinOS applications follow a small set of practical principles:

- Build focused applications around a clear user need.
- Reuse shared interface and policy components.
- Prefer open, vendorable technologies.
- Keep deployment simple and predictable.
- Treat accessibility, security, and plain language as default requirements.
- Store canonical content once and assemble it into different experiences.

The Marin App Shell

The Marin App Shell supplies common application structure and behavior. It establishes recognizable identity while leaving application-specific content and workflows under the application's control.

The shell can provide navigation, application identity, release status, standard informational links, managed icons, and common visual behavior.

Each application still owns its business rules, data handling, content, validation, and security decisions. Shared structure does not remove application accountability.

Single-source learning content

Marin Learn stores instructional narrative in reusable ContentUnits. A learning material, quiz, or slide deck references those units rather than maintaining an independent copy.

Output	How it uses canonical content
Reading view	Renders the full ordered material
Quiz	Links questions to the source content and objective
Slides	Uses a ContentUnit title, abstract, or full body

This does not prohibit presentation-specific information. Speaker notes, slide layouts, and question feedback have distinct purposes and remain separate.

Accessibility and security by default

Accessibility and security are enforced throughout the workflow rather than treated as optional cleanup.

- The editor restricts unsupported formatting.
- Images require meaningful alternative text or an explicit decorative decision.
- Heading structure and references are validated.
- Imported ZIP files, Markdown, links, and SVG assets are treated as untrusted input.
- Export is blocked while errors would make the package invalid or unsafe.

> *A stable renderer cannot compensate for unsafe or structurally invalid source content.*

A practical authoring workflow

1. Define what the learner should know or be able to do.
2. Create focused reusable ContentUnits.
3. Assemble the ContentUnits into a learning material.
4. Create questions that assess objectives and identify their source content.
5. Assemble slides from the same ContentUnits.
6. Resolve validation errors and review warnings.
7. Export the portable Marin Learn package.